Skill vs. Luck in Major Sports Leagues

IML Final Report, Fall 2025

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1 Introduction

In team sports, the outcome of a game typically depends not only on the relative strengths and skill levels of the teams involved, but also on elements of luck. In this project, we investigate how these factors affect the outcome of a game. We seek to answer questions such as the following: How can we quantify the relative influence of skill versus luck on game outcomes? How do major team sports compare in terms of the degree of skill involved in game outcomes? How has the relative share of skill versus luck evolved over time?

2 Description of Data

For this project we selected three US sports leagues and four European soccer leagues. The US leagues we chose are the National Football League (NFL), National Basketball Association (NBA), and Major League Baseball (MLB), and the European leagues are the Premier League (England), the Bundesliga (Germany), La Liga (Spain), and Serie A (Italy). Together, these seven leagues are among the most popular team sports leagues in the world, they represent a range of different sports that vary widely in terms of the lengths of their seasons and the way these seasons are structured.

Our skill vs. luck analysis is based on the game outcomes (win, loss, draw), along with a home/away team flag indicating which team in a game was the home team and which was the away team. The relevant data was sourced from the websites [1] (NBA data), [2] (NFL data), [4] (MLB data), and [3] (data for European soccer leagues). Table 1 summarizes the data we collected for our analysis.

League	Number of seasons	Number of games
NBA	76 (1949 - 2024)	66,410
MLB	45 (1980 - 2024)	101,135
NFL	59 (1966 - 2024)	13,500
Premier League	22 (2001 - 2024)	8,360
La Liga	23 (2000 - 2024)	8,740
Serie A	23 (2000 - 2024)	8,518
Bundesliga	21 (2004 - 2024)	6,426

Table 1: Summary of data used for our analysis.

3 Methodology

To quantify the relative influences of skill and luck on the outcomes of games, we proceed as follows.

We first introduce three metrics that all measure, though in very different ways, the influence of skill and luck on the outcomes of a game.

- **Win percentage dispersion.** This is the standard deviation of the set of win percentages across all seasons and teams. In a purely skill-based league with each team playing n games per season, this dispersion will be $\approx \sqrt{1/12}$. This is because in such a league, the strongest team will win all n of its games, the weakest team will win none of its games, while the number of wins for teams of intermediate strengths will cover the range $\{1, \ldots, n-1\}$ in an approximately uniform manner. Thus, the win percentages of the teams will be approximately uniformly distributed over the interval $[0, 1]$, and their standard deviation will be approximately that of a uniform distribution on $[0, 1]$, i.e., $\sqrt{1/12}$.

  By contrast, in a purely luck-based league, the number of wins will follow a binomial distribution with parameters n and $1/2$ and hence be approximately normally distributed with standard deviation proportional to $\sqrt{n}$. As a result, the standard deviation of win percentages will be proportional to $1/\sqrt{n}$, a much smaller quantity than that for a skill-based league.

- **First vs. second half season correlation.** This is defined as the Spearman correlation coefficient r between the win percentages in the first and second halves of the season for all teams in the league. For a purely skill-based league, there will be a strong positive correlation between these win percentages (i.e., $r \approx 1$), while for a purely luck-based league, there will be no correlation (i.e, $r \approx 0$).

- **Upset frequency.** The upset frequency is defined as the proportion of games in a season in which the higher-ranked team loses. In a purely skill-based league the higher ranked team always wins, so the upset frequency will be 0, whereas in a purely luck-based league, with game outcomes determined randomly, the upset frequency will be approximately 1/2.

As our next step, we created, for each league and season, two types of synthetic leagues that, in a sense, represent idealized versions of these leagues/seasons assuming either that all game outcomes are decided by skill alone, or that all game outcomes are decided by luck alone.

- **Skill-based synthetic league**: For each game, we declare the higher-ranked team based on the end-of-season rankings to be the winner.

- **Luck-based synthetic league**: For each game, we simulate the result by flipping a biased coin, where the probability for a home team to win is the overall home win rate in this sport league. (In the case of the soccer leagues, we use an analogous ternary coin to account for draws.)

The final step consists of translating, for each of the three metrics defined above, the raw value of that metric for the given league/season to a "skill share" and a "luck share", representing the relative contributions of skill and luck to the outcome of a game. To this end, we express the actual value of the metric as a weighted average of extremes for skill-based and luck-based synthetic leagues:

$$M = q \cdot M_S + (1 - q) \cdot M_L,$$

where M is the value of a metric in the actual league, M_S is the value of the same metric in the corresponding skill-based synthetic league, and M_L is the value of the same metric in the corresponding luck-based synthetic league. The weights q and $1 - q$ represent the shares of skill and luck with respect to the given metric.

4 Results

Figures 1 and 2 show the skill shares for the seven leagues, as measured by our first two metrics, win percentage dispersion and first vs. second half correlation. With respect to both of these metrics, the NBA has the highest skill share among the US leagues, followed by the NFL, while the MLB has the lowest skill share among those leagues. The skill shares for European leagues generally fall between those for the NBA and MLB, with the Premier League having the highest skill share among European leagues.

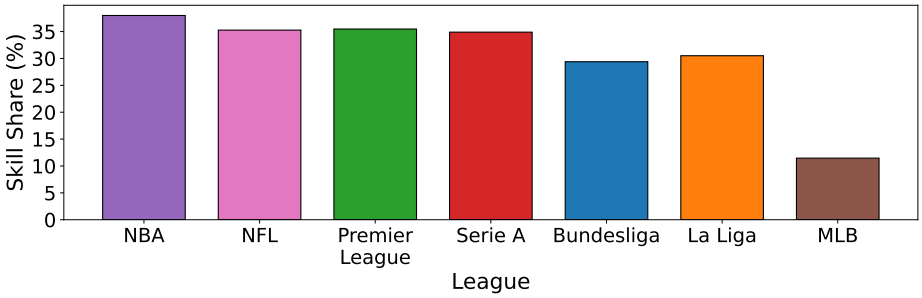


Figure 1: Skill share by league, as measured by win percentage dispersion.

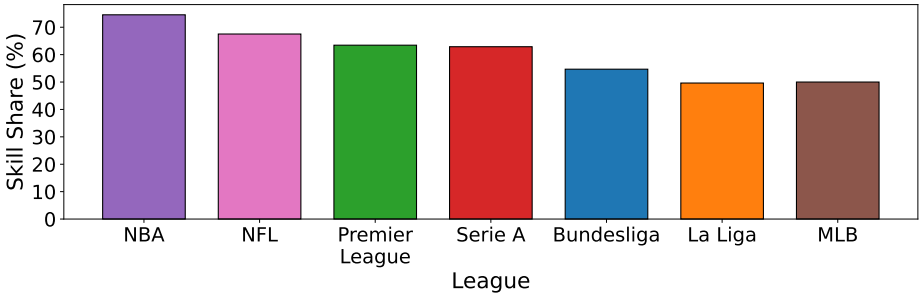


Figure 2: Skill share by league, as measured by first versus second half correlation.

Our third metric, the upset frequency, is based on a much larger data set (namely the set of all games in a season) than the other two metrics and thus allows a breakdown of skill shares by season. The results are shown in Figure 3. The figure confirms the overall trends observed in Figures 1 and 2, in that the NFL and MLB usually have the highest resp. lowest skill shares among all leagues, with the other leagues falling between these extremes. The figure also shows that there are substantial seasonal variations in the skill shares, suggesting that there is a significant amount of randomness in the factors that determine skill shares. However, there does not seem to be a noticeable upward or downward trend in skill shares within the period covered in the figure.

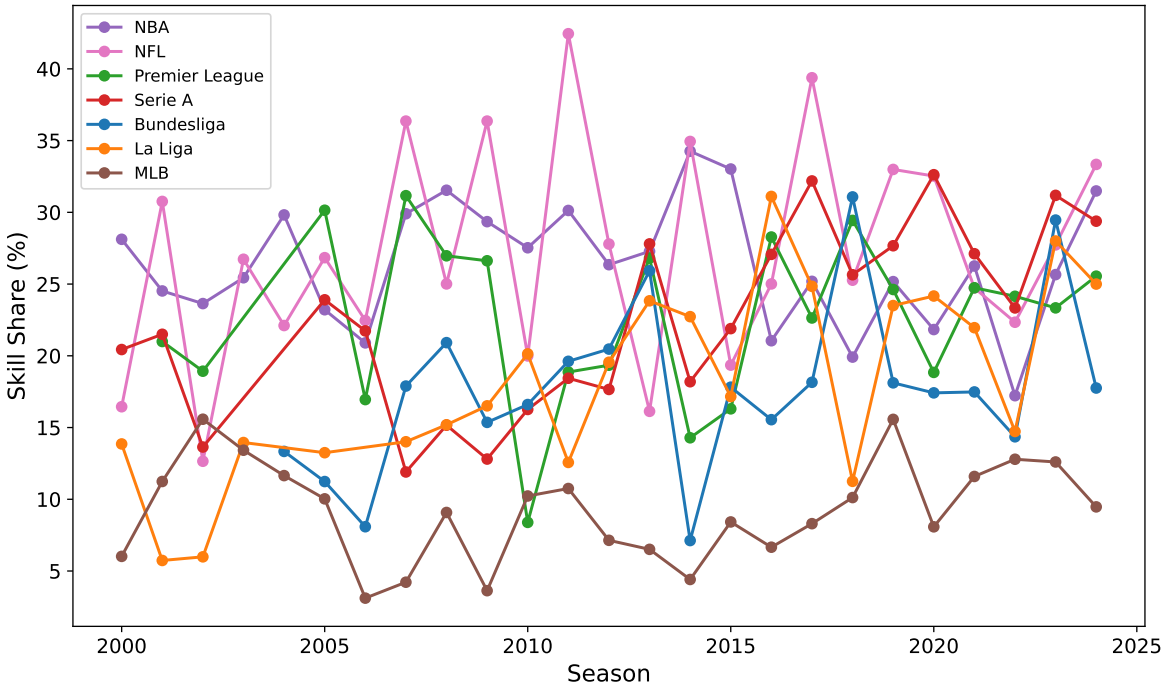


Figure 3: Skill share by league and season, as measured by upset frequency.

References

[1] Basketball-Reference.com. NBA league statistics and history. `www.basketball-reference.com/leagues/`, 2025.

[2] Toby Crabtree. NFL scores and betting data. `www.kaggle.com/datasets/tobycrabtree/nfl-scores-and-betting-data`, 2020.

[3] Adam Gábor. Club Football Match Data. `https://github.com/xgabora/Club-Football-Match-Data-2000-2025`, 2025.

[4] Retrosheet. Retrosheet game logs. `www.retrosheet.org/gamelogs/`, 2024.

Github Repository. Further details about this project can be found at the project's Github repository, `https://github.com/mikeyautorino2/skillvsluck`. The repository includes all data sources, code, visualizations, and league-level data sets used throughout the project. It also includes documentation and reproducible workflows used to generate the analyses and figures presented in this report.